SQL Intern Assignment: E-Commerce Database Project

Overview

In this assignment, you will work with a fictional e-commerce database called "TechMart." You'll set up the database, create tables, insert sample data, and then solve various SQL challenges that test your knowledge of SQL concepts.

Part 1: Database Setup and Sample Data

Step 1: Create the Database

```
-- Create the TechMart database
CREATE DATABASE TechMart;


-- Use the TechMart database
USE TechMart;
```

Step 2: Create Tables

```
-- Customers table
CREATE TABLE Customers (
  customer_id INT PRIMARY KEY,
  first_name VARCHAR(50) NOT NULL,
  last_name VARCHAR(50) NOT NULL,
  email VARCHAR(100) UNIQUE NOT NULL,
  phone VARCHAR(20),
  address VARCHAR(200),
  city VARCHAR(50),
  state VARCHAR(50),
  zip_code VARCHAR(20),
  registration_date DATETIME DEFAULT CURRENT_TIMESTAMP,
```

```
  loyalty_points INT DEFAULT 0
);

-- Product Categories table
CREATE TABLE ProductCategories (
  category_id INT PRIMARY KEY,
  category_name VARCHAR(50) NOT NULL,
  description TEXT,
  parent_category_id INT,
  FOREIGN KEY (parent_category_id) REFERENCES ProductCategories(category_id)
);

-- Products table
CREATE TABLE Products (
  product_id INT PRIMARY KEY,
  product_name VARCHAR(100) NOT NULL,
  description TEXT,
  category_id INT,
  price DECIMAL(10, 2) NOT NULL,
  stock_quantity INT DEFAULT 0,
  manufacturer VARCHAR(100),
  release_date DATE,
  is_active BOOLEAN DEFAULT TRUE,
  FOREIGN KEY (category_id) REFERENCES ProductCategories(category_id)
);
```

```sql
-- Orders table
CREATE TABLE Orders (
  order_id INT PRIMARY KEY,
  customer_id INT,
  order_date DATETIME DEFAULT CURRENT_TIMESTAMP,
  total_amount DECIMAL(12, 2) NOT NULL,
  shipping_address VARCHAR(200),
  shipping_city VARCHAR(50),
  shipping_state VARCHAR(50),
  shipping_zip_code VARCHAR(20),
  status VARCHAR(50) DEFAULT 'Pending',
  payment_method VARCHAR(50),
  FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)
);

-- Order Items table (junction table between Orders and Products)
CREATE TABLE OrderItems (
  order_item_id INT PRIMARY KEY,
  order_id INT,
  product_id INT,
  quantity INT NOT NULL,
  unit_price DECIMAL(10, 2) NOT NULL,
  discount DECIMAL(10, 2) DEFAULT 0.00,
  FOREIGN KEY (order_id) REFERENCES Orders(order_id),
  FOREIGN KEY (product_id) REFERENCES Products(product_id)
);
```

```sql
-- Employees table
CREATE TABLE Employees (
  employee_id INT PRIMARY KEY,
  first_name VARCHAR(50) NOT NULL,
  last_name VARCHAR(50) NOT NULL,
  email VARCHAR(100) UNIQUE NOT NULL,
  phone VARCHAR(20),
  hire_date DATE NOT NULL,
  job_title VARCHAR(100),
  department VARCHAR(100),
  manager_id INT,
  salary DECIMAL(10, 2),
  FOREIGN KEY (manager_id) REFERENCES Employees(employee_id)
);

-- Reviews table
CREATE TABLE Reviews (
  review_id INT PRIMARY KEY,
  product_id INT,
  customer_id INT,
  rating INT CHECK (rating BETWEEN 1 AND 5),
  comment TEXT,
  review_date DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (product_id) REFERENCES Products(product_id),
  FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)
);
```

```
-- Promotions table
CREATE TABLE Promotions (
  promotion_id INT PRIMARY KEY,
  promotion_name VARCHAR(100) NOT NULL,
  description TEXT,
  discount_percentage DECIMAL(5, 2),
  start_date DATE NOT NULL,
  end_date DATE NOT NULL,
  is_active BOOLEAN DEFAULT TRUE
);

-- Product Promotions mapping (junction table between Products and Promotions)
CREATE TABLE ProductPromotions (
  product_id INT,
  promotion_id INT,
  PRIMARY KEY (product_id, promotion_id),
  FOREIGN KEY (product_id) REFERENCES Products(product_id),
  FOREIGN KEY (promotion_id) REFERENCES Promotions(promotion_id)
);
```

Step 3: Insert Sample Data

```
-- Insert data into Customers table
INSERT INTO Customers (customer_id, first_name, last_name, email, phone, address, city, state, zip_code, registration_date, loyalty_points)
VALUES
  (1, 'John', 'Smith', 'john.smith@email.com', '555-123-4567', '123 Main St', 'Austin', 'TX', '78701', '2023-01-15', 250),
```

(2, 'Emma', 'Johnson', 'emma.j@email.com', '555-234-5678', '456 Elm St', 'Seattle', 'WA', '98101', '2023-02-20', 175),

(3, 'Michael', 'Williams', 'michael.w@email.com', '555-345-6789', '789 Oak Ave', 'Chicago', 'IL', '60601', '2023-03-10', 320),

(4, 'Sophia', 'Brown', 'sophia.b@email.com', '555-456-7890', '101 Pine Rd', 'New York', 'NY', '10001', '2023-04-05', 100),

(5, 'Robert', 'Jones', 'robert.j@email.com', '555-567-8901', '202 Maple Dr', 'Los Angeles', 'CA', '90001', '2023-04-12', 280),

(6, 'Olivia', 'Garcia', 'olivia.g@email.com', '555-678-9012', '303 Birch Ln', 'Miami', 'FL', '33101', '2023-05-18', 150),

(7, 'William', 'Miller', 'william.m@email.com', '555-789-0123', '404 Cedar Ct', 'Denver', 'CO', '80201', '2023-06-22', 90),

(8, 'Ava', 'Davis', 'ava.d@email.com', '555-890-1234', '505 Walnut Pl', 'Portland', 'OR', '97201', '2023-07-14', 210),

(9, 'James', 'Rodriguez', 'james.r@email.com', '555-901-2345', '606 Spruce Way', 'Boston', 'MA', '02101', '2023-08-08', 170),

(10, 'Isabella', 'Martinez', 'isabella.m@email.com', '555-012-3456', '707 Fir Blvd', 'Phoenix', 'AZ', '85001', '2023-09-19', 300),

(11, 'David', 'Anderson', 'david.a@email.com', '555-321-7654', '808 Cypress St', 'San Francisco', 'CA', '94101', '2024-01-10', 220),

(12, 'Mia', 'Taylor', 'mia.t@email.com', '555-432-8765', '909 Redwood Dr', 'Atlanta', 'GA', '30301', '2024-02-05', 180);

```
-- Insert data into ProductCategories table

INSERT INTO ProductCategories (category_id, category_name, description,
parent_category_id)
```

VALUES

(1, 'Electronics', 'Electronic devices and accessories', NULL),

(2, 'Computers', 'Desktop and laptop computers', 1),

(3, 'Smartphones', 'Mobile phones and accessories', 1),

(4, 'Audio', 'Headphones, speakers, and audio equipment', 1),

(5, 'Gaming', 'Video games and gaming equipment', NULL),

(6, 'Consoles', 'Gaming consoles and accessories', 5),

(7, 'PC Gaming', 'PC gaming hardware and accessories', 5),

(8, 'Home Appliances', 'Household electronic appliances', NULL),

(9, 'Kitchen', 'Kitchen appliances and equipment', 8);

```
-- Insert data into Products table
INSERT INTO Products (product_id, product_name, description, category_id,
price, stock_quantity, manufacturer, release_date, is_active)
```

VALUES

(101, 'UltraBook Pro', '15-inch laptop with high performance', 2, 1299.99, 50, 'TechCorp', '2023-03-15', TRUE),

(102, 'SmartPhone X', 'Latest smartphone with advanced features', 3, 899.99, 120, 'MobiTech', '2023-05-20', TRUE),

(103, 'Noise Cancelling Headphones', 'Premium headphones with noise cancellation', 4, 249.99, 85, 'AudioMax', '2023-02-10', TRUE),

(104, 'Gaming Console Pro', 'Next-gen gaming console', 6, 499.99, 30, 'GameTech', '2023-11-05', TRUE),

(105, 'Smart Watch', 'Fitness and notification tracker', 1, 199.99, 75, 'MobiTech', '2023-07-14', TRUE),

(106, 'Bluetooth Speaker', 'Portable bluetooth speaker with 20hr battery', 4, 129.99, 60, 'AudioMax', '2023-08-22', TRUE),

(107, 'Desktop Computer', 'High-performance desktop for professionals', 2, 1599.99, 25, 'TechCorp', '2023-09-30', TRUE),

(108, 'Tablet Pro', '12-inch tablet with stylus support', 1, 699.99, 40, 'MobiTech', '2023-10-15', TRUE),

(109, 'Wireless Gaming Mouse', 'High precision gaming mouse', 7, 89.99, 100, 'GameTech', '2023-04-18', TRUE),

(110, 'Coffee Maker', 'Programmable coffee maker with timer', 9, 149.99, 35, 'HomeElectronics', '2023-06-25', TRUE),

(111, 'Microwave Oven', 'Digital microwave with multiple cooking modes', 9, 179.99, 30, 'HomeElectronics', '2023-07-05', TRUE),

(112, 'Gaming Keyboard', 'Mechanical RGB gaming keyboard', 7, 129.99, 45, 'GameTech', '2023-08-10', TRUE),

(113, 'Wireless Earbuds', 'True wireless earbuds with charging case', 4, 159.99, 90, 'AudioMax', '2023-09-18', TRUE),

(114, 'External Hard Drive', '2TB portable external hard drive', 2, 119.99, 70, 'DataStore', '2023-05-28', TRUE),

(115, 'Smartphone Y', 'Mid-range smartphone with excellent camera', 3, 599.99, 65, 'MobiTech', '2023-10-25', TRUE),

(116, 'Ultra HD Monitor', '27-inch 4K monitor for professionals', 2, 349.99, 55, 'ViewTech', '2023-11-15', TRUE),

(117, 'Smart Blender', 'Programmable blender with multiple speeds', 9, 89.99, 40, 'HomeElectronics', '2023-12-10', FALSE),

(118, 'Gaming Headset', 'Surround sound gaming headset with mic', 7, 149.99, 60, 'GameTech', '2024-01-05', TRUE),

(119, 'Wireless Router', 'High-speed dual-band wireless router', 1, 129.99, 50, 'NetConnect', '2024-02-12', TRUE),

(120, 'Portable Power Bank', '20000mAh fast charging power bank', 3, 49.99, 80, 'PowerTech', '2024-03-20', TRUE);

```
-- Insert data into Orders table
INSERT INTO Orders (order_id, customer_id, order_date, total_amount,
shipping_address, shipping_city, shipping_state, shipping_zip_code,
status, payment_method)
```

VALUES

(1001, 1, '2023-05-10 14:30:00', 1549.98, '123 Main St', 'Austin', 'TX', '78701', 'Delivered', 'Credit Card'),

(1002, 2, '2023-05-15 10:15:00', 899.99, '456 Elm St', 'Seattle', 'WA', '98101', 'Delivered', 'PayPal'),

(1003, 3, '2023-06-02 09:45:00', 379.98, '789 Oak Ave', 'Chicago', 'IL', '60601', 'Delivered', 'Credit Card'),

(1004, 4, '2023-06-10 16:20:00', 499.99, '101 Pine Rd', 'New York', 'NY', '10001', 'Delivered', 'Debit Card'),

(1005, 5, '2023-07-05 11:30:00', 329.98, '202 Maple Dr', 'Los Angeles', 'CA', '90001', 'Delivered', 'Credit Card'),

(1006, 1, '2023-07-20 14:45:00', 699.99, '123 Main St', 'Austin', 'TX', '78701', 'Delivered', 'Credit Card'),

(1007, 6, '2023-08-08 13:10:00', 449.98, '303 Birch Ln', 'Miami', 'FL', '33101', 'Delivered', 'PayPal'),

(1008, 7, '2023-08-25 15:25:00', 1599.99, '404 Cedar Ct', 'Denver', 'CO', '80201', 'Shipped', 'Credit Card'),

(1009, 8, '2023-09-12 10:05:00', 249.99, '505 Walnut Pl', 'Portland', 'OR', '97201', 'Delivered', 'Debit Card'),

(1010, 9, '2023-09-30 17:40:00', 219.98, '606 Spruce Way', 'Boston', 'MA', '02101', 'Delivered', 'Credit Card'),

(1011, 10, '2023-10-15 09:30:00', 1099.98, '707 Fir Blvd', 'Phoenix', 'AZ', '85001', 'Shipped', 'PayPal'),

(1012, 11, '2023-11-05 14:15:00', 679.97, '808 Cypress St', 'San Francisco', 'CA', '94101', 'Processing', 'Credit Card'),

(1013, 12, '2023-11-20 11:50:00', 279.98, '909 Redwood Dr', 'Atlanta', 'GA', '30301', 'Shipped', 'Debit Card'),

(1014, 3, '2023-12-10 16:05:00', 849.98, '789 Oak Ave', 'Chicago', 'IL', '60601', 'Delivered', 'Credit Card'),

(1015, 5, '2023-12-28 13:40:00', 399.98, '202 Maple Dr', 'Los Angeles', 'CA', '90001', 'Delivered', 'PayPal'),

(1016, 2, '2024-01-15 10:25:00', 1129.98, '456 Elm St', 'Seattle', 'WA', '98101', 'Shipped', 'Credit Card'),

(1017, 8, '2024-02-02 15:10:00', 459.98, '505 Walnut Pl', 'Portland', 'OR', '97201', 'Processing', 'Debit Card'),

(1018, 1, '2024-02-20 09:55:00', 349.99, '123 Main St', 'Austin', 'TX', '78701', 'Processing', 'Credit Card'),

(1019, 4, '2024-03-08 12:30:00', 279.98, '101 Pine Rd', 'New York', 'NY', '10001', 'Pending', 'PayPal'),

(1020, 6, '2024-03-25 14:20:00', 549.98, '303 Birch Ln', 'Miami', 'FL', '33101', 'Pending', 'Credit Card');

```
-- Insert data into OrderItems table

INSERT INTO OrderItems (order_item_id, order_id, product_id, quantity,
unit_price, discount)

VALUES

  (2001, 1001, 101, 1, 1299.99, 0.00),

  (2002, 1001, 103, 1, 249.99, 0.00),

  (2003, 1002, 102, 1, 899.99, 0.00),

  (2004, 1003, 103, 1, 249.99, 0.00),

  (2005, 1003, 106, 1, 129.99, 0.00),

  (2006, 1004, 104, 1, 499.99, 0.00),

  (2007, 1005, 105, 1, 199.99, 0.00),

  (2008, 1005, 106, 1, 129.99, 0.00),

  (2009, 1006, 108, 1, 699.99, 0.00),

  (2010, 1007, 109, 1, 89.99, 0.00),

  (2011, 1007, 113, 1, 159.99, 0.00),

  (2012, 1007, 110, 1, 149.99, 0.00),

  (2013, 1008, 107, 1, 1599.99, 0.00),

  (2014, 1009, 103, 1, 249.99, 0.00),

  (2015, 1010, 109, 1, 89.99, 0.00),

  (2016, 1010, 120, 1, 49.99, 0.00),

  (2017, 1010, 114, 1, 119.99, 40.00),

  (2018, 1011, 115, 1, 599.99, 0.00),

  (2019, 1011, 116, 1, 349.99, 0.00),

  (2020, 1011, 119, 1, 129.99, 0.00),

  (2021, 1012, 118, 1, 149.99, 0.00),

  (2022, 1012, 116, 1, 349.99, 0.00),

  (2023, 1012, 120, 2, 49.99, 0.00),

  (2024, 1013, 113, 1, 159.99, 0.00),
```

(2025, 1013, 120, 1, 49.99, 0.00),

(2026, 1013, 109, 1, 89.99, 20.00),

(2027, 1014, 102, 1, 899.99, 50.00),

(2028, 1015, 110, 1, 149.99, 0.00),

(2029, 1015, 111, 1, 179.99, 0.00),

(2030, 1015, 120, 1, 49.99, 0.00),

(2031, 1016, 101, 1, 1299.99, 200.00),

(2032, 1016, 109, 1, 89.99, 0.00),

(2033, 1016, 120, 1, 49.99, 0.00),

(2034, 1017, 118, 1, 149.99, 0.00),

(2035, 1017, 113, 1, 159.99, 0.00),

(2036, 1017, 120, 3, 49.99, 0.00),

(2037, 1018, 116, 1, 349.99, 0.00),

(2038, 1019, 113, 1, 159.99, 0.00),

(2039, 1019, 120, 1, 49.99, 0.00),

(2040, 1019, 106, 1, 129.99, 60.00),

(2041, 1020, 104, 1, 499.99, 0.00),

(2042, 1020, 120, 1, 49.99, 0.00);

```
-- Insert data into Employees table

INSERT INTO Employees (employee_id, first_name, last_name, email, phone,
hire_date, job_title, department, manager_id, salary)
```

VALUES

(501, 'Thomas', 'Wilson', 'thomas.w@techmart.com', '555-111-2222', '2022-01-15', 'CEO', 'Executive', NULL, 250000.00),

(502, 'Sarah', 'Johnson', 'sarah.j@techmart.com', '555-222-3333', '2022-02-20', 'CTO', 'Executive', 501, 200000.00),

(503, 'Daniel', 'Brown', 'daniel.b@techmart.com', '555-333-4444', '2022-03-10', 'CFO', 'Executive', 501, 200000.00),

(504, 'Jessica', 'Smith', 'jessica.s@techmart.com', '555-444-5555', '2022-04-05', 'IT Manager', 'IT', 502, 120000.00),

(505, 'Andrew', 'Davis', 'andrew.d@techmart.com', '555-555-6666', '2022-05-12', 'Sales Manager', 'Sales', 501, 110000.00),

(506, 'Lisa', 'Miller', 'lisa.m@techmart.com', '555-666-7777', '2022-06-18', 'Marketing Manager', 'Marketing', 501, 110000.00),

(507, 'Robert', 'Taylor', 'robert.t@techmart.com', '555-777-8888', '2022-07-22', 'HR Manager', 'HR', 501, 105000.00),

(508, 'Emily', 'Anderson', 'emily.a@techmart.com', '555-888-9999', '2022-08-14', 'IT Specialist', 'IT', 504, 85000.00),

(509, 'Michael', 'Clark', 'michael.c@techmart.com', '555-999-0000', '2022-09-08', 'IT Specialist', 'IT', 504, 85000.00),

(510, 'Amanda', 'Wright', 'amanda.w@techmart.com', '555-000-1111', '2022-10-19', 'Sales Representative', 'Sales', 505, 65000.00),

(511, 'Joshua', 'Garcia', 'joshua.g@techmart.com', '555-111-3333', '2022-11-10', 'Sales Representative', 'Sales', 505, 65000.00),

(512, 'Melissa', 'Martinez', 'melissa.m@techmart.com', '555-222-4444', '2022-12-05', 'Marketing Specialist', 'Marketing', 506, 70000.00),

(513, 'Christopher', 'Lee', 'christopher.l@techmart.com', '555-333-5555', '2023-01-15', 'HR Specialist', 'HR', 507, 65000.00),

(514, 'Nicole', 'Harris', 'nicole.h@techmart.com', '555-444-6666', '2023-02-20', 'IT Support', 'IT', 504, 60000.00),

(515, 'Brian', 'Young', 'brian.y@techmart.com', '555-555-7777', '2023-03-18', 'Sales Representative', 'Sales', 505, 65000.00);

```
-- Insert data into Reviews table

INSERT INTO Reviews (review_id, product_id, customer_id, rating, comment, review_date)
```

VALUES

(3001, 101, 1, 5, 'Excellent laptop, very fast and reliable', '2023-05-20 15:30:00'),

(3002, 102, 2, 4, 'Great phone but battery life could be better', '2023-05-25 10:45:00'),

(3003, 103, 3, 5, 'Amazing sound quality, best headphones I\'ve owned', '2023-06-10 09:20:00'),

(3004, 104, 4, 3, 'Good console but lacks some features compared to competitors', '2023-06-18 14:10:00'),

(3005, 105, 5, 4, 'Nice smart watch, tracks fitness well', '2023-07-12 11:25:00'),

(3006, 106, 6, 5, 'Incredible sound for such a small speaker', '2023-07-28 16:40:00'),

(3007, 107, 7, 4, 'Powerful desktop, great for my work needs', '2023-09-05 13:15:00'),

(3008, 108, 8, 5, 'The tablet is perfect for my design work', '2023-09-20 10:30:00'),

(3009, 109, 9, 4, 'Responsive gaming mouse, comfortable to use', '2023-10-08 17:50:00'),

(3010, 110, 10, 5, 'Makes great coffee and easy to program', '2023-10-25 09:10:00'),

(3011, 111, 11, 3, 'Works well but a bit noisy', '2023-11-12 12:45:00'),

(3012, 112, 12, 5, 'Keys feel great, RGB lighting is awesome', '2023-11-30 15:20:00'),

(3013, 113, 1, 4, 'Good sound quality but fit is a bit loose', '2023-12-15 14:35:00'),

(3014, 101, 3, 4, 'Good performance but runs hot under heavy load', '2023-12-28 11:55:00'),

(3015, 102, 5, 5, 'Camera quality is exceptional!', '2024-01-10 10:15:00'),

(3016, 104, 7, 4, 'Great gaming experience', '2024-01-25 16:30:00'),

(3017, 105, 9, 3, 'Battery life is shorter than advertised', '2024-02-08 13:40:00'),

(3018, 108, 11, 5, 'Perfect for both work and entertainment', '2024-02-22 09:25:00'),

(3019, 113, 2, 4, 'Sound quality is great, battery life could be better', '2024-03-10 14:50:00'),

(3020, 116, 4, 5, 'Amazing display quality, perfect for my graphic design work', '2024-03-28 11:05:00');

```
-- Insert data into Promotions table

INSERT INTO Promotions (promotion_id, promotion_name, description,
discount_percentage, start_date, end_date, is_active)
```

VALUES

(401, 'Summer Sale', 'Annual summer discount event', 15.00, '2023-06-01', '2023-06-30', FALSE),

(402, 'Back to School', 'Discounts on electronics for students', 10.00, '2023-08-01', '2023-08-31', FALSE),

(403, 'Black Friday', 'Biggest sale of the year', 25.00, '2023-11-24', '2023-11-27', FALSE),

(404, 'Holiday Season', 'End of year holiday promotions', 20.00, '2023-12-15', '2023-12-31', FALSE),

(405, 'New Year Sale', 'Start the year with savings', 15.00, '2024-01-01', '2024-01-15', FALSE),

(406, 'Winter Clearance', 'Clearing inventory for new products', 30.00, '2024-02-01', '2024-02-28', FALSE),

(407, 'Spring Tech Fest', 'Discounts on latest technology products', 12.00, '2024-03-15', '2024-04-15', TRUE);

```
-- Insert data into ProductPromotions table
INSERT INTO ProductPromotions (product_id, promotion_id)
```

VALUES

(101, 402), -- UltraBook Pro in Back to School promotion

(102, 401), -- SmartPhone X in Summer Sale

(103, 401), -- Noise Cancelling Headphones in Summer Sale

(104, 403), -- Gaming Console Pro in Black Friday

(105, 401), -- Smart Watch in Summer Sale

(106, 402), -- Bluetooth Speaker in Back to School

(107, 403), -- Desktop Computer in Black Friday

(108, 404), -- Tablet Pro in Holiday Season

(109, 402), -- Wireless Gaming Mouse in Back to School

(110, 404), -- Coffee Maker in Holiday Season

(111, 405), -- Microwave Oven in New Year Sale

(112, 403), -- Gaming Keyboard in Black Friday

(113, 402), -- Wireless Earbuds in Back to School

(114, 401), -- External Hard Drive in Summer Sale

(115, 404), -- Smartphone Y in Holiday Season

(116, 405), -- Ultra HD Monitor in New Year Sale

(118, 406), -- Gaming Headset in Winter Clearance

(119, 406), -- Wireless Router in Winter Clearance

(120, 407), -- Portable Power Bank in Spring Tech Fest

(101, 407), -- UltraBook Pro in Spring Tech Fest

(102, 407), -- SmartPhone X in Spring Tech Fest

(103, 407), -- Noise Cancelling Headphones in Spring Tech Fest

(104, 407); -- Gaming Console Pro in Spring Tech Fest

Part 2: SQL Challenges

Complete the following 25 questions using the TechMart database:

1. Display all products that cost more than $500 and have at least 50 items in stock, sorted by price from highest to lowest.

2. Find all customers who have registered in 2023 and have more than 200 loyalty points. Show their full name as a single column, email, and city.

3. Update the loyalty points for customers in Texas by adding 50 points to their current total.

4. Create a query to identify products that need restocking (stock_quantity less than 30).

5. Calculate the average price of products in each category, but only include categories where the average price is greater than $200.

6. Find all customers who haven't placed any orders yet.

7. Display the top 5 customers who have spent the most money on purchases, showing their name and total amount spent.

8. For each product, show its name, price, and the average rating it has received.

9. Find all products that were ordered in February 2024, along with the customers who ordered them.

10. List all employees along with their direct manager's name (if they have one).

11. Find all products that have never been reviewed by any customer.

12. Identify customers who have purchased products from at least 3 different product categories.

13. Find all products that are not currently part of any active promotion.

14. List customers who have purchased the same product more than once across different orders.

15. Find departments with more than 3 employees, showing the department name and the number of employees.

16. Calculate the total revenue generated for each month in 2023, sorted chronologically.

17. Find the most popular product category based on the number of orders placed.

18. Identify the employees who earn a salary higher than the average salary of their department.

19. Calculate the percentage of reviews that are 4 stars or higher for each product that has at least 2 reviews.

20. Find the average order value for each payment method, and include the number of orders placed with each method.

21. Create a query to analyze which day of the week has the highest number of orders.

22. Generate a report showing the quarterly sales trends for 2023, broken down by product category.

23. Create a view that displays a summary of each customer's purchase history, including their most purchased category and total spend.

24. Calculate the employee hierarchy depth, showing each employee's level in the organization (CEO being level 1).

25. Analyze the correlation between product price and customer review ratings. Do more expensive products tend to get higher ratings?

Submission Guidelines:-

To successfully complete and submit this assignment, please follow the steps below:

1. Write SQL Queries:
    - Write correct SQL queries for all 25 questions provided in the assignment using MySQL Workbench.
    - Ensure the queries are properly commented and organized for readability.
2. Export the Script:
    - Once you've written and tested all queries, export the entire SQL script file from MySQL Workbench.
    - Save the file in .sql format (e.g., YourName_TechMart_Assignment.sql).
3. Capture Execution Logs:
    - Run each query in MySQL Workbench.
    - Take 5 to 8 clear screenshots of the output/result/log section while executing the queries.
    - Include both successful query execution and meaningful results wherever applicable.
4. Create a PDF Report:
    - Paste the captured screenshots into a Word or Google Docs file.
    - Optionally, add brief captions or titles to indicate what each screenshot represents.

- Export the document as a PDF (e.g., YourName_TechMart_Logs.pdf).

5. Final Submission:
    - Submit the following two files:
        - The exported SQL script file (.sql)
        - The PDF report with screenshots (.pdf)
6. Deadline:
    - Submit both files before the due date as instructed by admin team.